

Introduction to Probability

CSU, Chico Math 314

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outline

Sample Space and Events

Set Operations

Probability

Sets and Probability

Conditional Probability

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- Law of Total Probability

- Bayes' Theorem

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Sample Space

Probability is interested in outcomes of experiments (observational studies, games, ...).

sample space

A sample space is the set of all possible outcomes of an experiment, commonly denoted $\mathcal{S}$.

event

An event is any subset of the sample space.

Sample Space, examples

Here are some examples of sample spaces:

- ▶ two websites that are classified as defective or non-defective
 - ▶ $\{NN, DN, ND, DD\}$
- ▶ a survey about people's expectations for the time (in months, say) until the next version of R releases.
 - ▶ $\{0, 1, 2, \dots\}$

Event, examples

Here are some examples of events corresponding to the sample spaces above:

- ▶ defective (D) or non-defective (N) websites
 - ▶ both work $== \{NN\}$, neither works $== \{DD\}$, **exactly** one works $== \{ND, DN\}$
- ▶ expectation for time until next version of R in months
 - ▶ $\{3\}$ or $\{0, 1, 2\}$

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Set Operations

Here's a few official definitions for some set operations:

union

The union of k sets A_j is the event consisting of A_1 or (not xor) A_2 or $\dots A_k$.

Mathematically, we write $\cup_{j=1}^k A_j$.

This is also thought of as **at least one of A_j happens**.

intersection

The intersection of k sets A_j is the event consisting of A_1 and A_2 and $\dots A_k$.

Mathematically, we write $\cap_{j=1}^k A_j$.

This is also thought of as **all of $A_1, \dots, A_k$ happen**.

Set Operations

complement

The complement A^c of A is the set of events that **are not in** A .

difference

The difference $B - A$ is the set of events **contained in** B **but not in** A .

Set Operations

disjoint

Two sets A and B are disjoint (or mutually exclusive) if A and B do not overlap, i.e. **have no events in common**.

Mathematically, we write $A \cap B = \emptyset$.

subset

The set A is said to be a subset of B if **all events in A imply an event of the set B** .

Mathematically, we write $A \subseteq B$.

We also write $a \in A$ **implies** $a \in B$.

Subsets

Some notes on subsets:

- ▶ Sometimes people will use the symbol $\subset$ to explicitly state that A is a subset of B , but certainly not equal to B .
- ▶ To show that $A = B$, one needs to show both that $x \in A$ implies $x \in B$ and that $x \in B$ implies $x \in A$.

Laws of Set Operations

There are well established laws of sets, analogous to laws of mathematics.

- ▶ Commutative: $A \cup B = B \cup A$ and
 $A \cap B = B \cap A$.
- ▶ Associative: $(A \cup B) \cup C = A \cup (B \cup C)$ and
 $(A \cap B) \cap C = A \cap (B \cap C)$.
- ▶ Distributive: $C \cap (A \cup B) = (C \cap A) \cup (C \cap B)$ and
 $C \cup (A \cap B) = (C \cup A) \cap (C \cup B)$.
- ▶ De Morgan's Laws: $(A \cup B)^c = A^c \cap B^c$ and
 $(A \cap B)^c = A^c \cup B^c$.

Disclaimer

Set operations, laws of set operations, and/or any combination of these don't come up very often. HOWEVER, having some set logic at your disposal at a future career you have just might come in handy when you need it.

It's also good to be familiar with R's syntax for working with set operations, particularly when considering massive data sets.

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Probability of Events

NOTE: The URL must be read in as a single line.

```
url <- "https://raw.githubusercontent.com/njlytal/  
MATH314/main/email.csv"  
email <- read.csv(url)
```

Let's check out spam and format.

Probability of Events

Here are some columns from the dataset `email`

	spam	num_char	format	number
47	0	8.72	0	small
24	0	0.20	0	small
30	0	2.21	1	small
41	0	12.27	0	small
33	0	10.61	0	small
25	0	4.94	1	none

Table 1: Four rows of data from the email data set

Probability of Events

What is the probability a randomly chosen email is marked as spam, $P(\text{spam})$?

```
mean(email[, "spam"])  
## [1] 0.09359857
```

Probability of Events

What is the probability a randomly chosen email is written in HTML, $P(\text{format})$?

Probability of Events

What is the probability a randomly chosen email is written in HTML **or** it's spam, $P(?)$?

Probability of Events

What is the probability a randomly chosen email is not written in HTML **and** it's not spam, $P(?)$?

Useful R Operators

For statements like the above, there are useful ways to create logical statements in R:

AND: &

OR: |

NOT: !

Examples:

```
mean(email$spam & email$format) # spam AND HTML
```

```
## [1] 0.04029584
```

```
mean(!email$spam) # NOT spam
```

```
## [1] 0.9064014
```

```
mean(!email$spam | !email$format)
```

```
# either NOT spam OR NOT HTML
```

```
## [1] 0.9597042
```

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Connecting Sets and Probability

Probability is formally defined as an abstract mathematical operation that maps sets to the interval $[0, 1]$. If we let $\mathcal{E}$ be the set of all possible events of $\mathcal{S}$, then mathematically we write $P : \mathcal{E} \rightarrow [0, 1]$.

More tangibly:

- ▶ $P(H) = 0.5$ says that the probability of flipping a head is $1/2$.
- ▶ much of the rest of this class is talking about the probability of different events, E or H or anything else.

Equally Likely Outcomes

The most common probability models are those consisting of finite sample spaces, where each of N outcomes is equally likely to occur:

- ▶ coin flip: $\mathcal{S} = \{H, T\}$, $P(H) = 1/2$
- ▶ die roll: $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$, $P(\text{roll } 1) = 1/6$
- ▶ If N elements in $\mathcal{S}$, then $P(E) = N(E)/N$

Probability, example

Imagine you roll two (fair) dice. What is the probability of the event $E = \{R_1 + R_2 = 7\}$?

- ▶ What is the sample space?
 - ▶ $\mathcal{S} = \{(1, 1), (1, 2), \dots, (6, 6)\}$ implies $N = 36$.
- ▶ How many ways can we roll a sum of 7?
 - ▶ $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$ implies $N(E) = 6$.
- ▶ The elements of E are disjoint and equally likely
- ▶ Answer: $P(E) = 1/6$

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Motivating Conditional Probability

Wouldn't it be great if there was a formal method to update probabilities given new information? Consider this scenario. You draw a card from a deck of 52 cards **and then** learn that the card you drew (still face down) is a face card. What is the probability that this card is a king?

- ▶ There's 12 face cards.
- ▶ 4 of the 12 face cards are kings.
- ▶ Therefore, there's a $1/3$ probability that the card you drew is a king.

Motivating Conditional Probability

Without any information that the card you drew is a face card, the probability of drawing a king is $4/52 = 0.08$. But if you do for a fact know that the card you have drawn is a face card, the probability of it being a king is $1/3 = 0.33$.

This new probability is **conditional** on knowing whether you have drawn a face card.

Conditional Probability

Conditional probability allows us to update probabilities given a new set of information; e.g. we learned that the card we drew was a face card.

conditional probability

For any two events A and B

$$P(B|A) = \frac{P(A \cap B)}{P(A)},$$

so long as $P(A) > 0$.

Conditional Probability, example

Let $P(A)$ be the probability that we draw a face card and let $P(B)$ be the probability that we draw a king.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(\text{face card and king})}{P(\text{face card})} = \frac{4/52}{12/52}.$$

Conditional Probability, example 2

Suppose you are working with a traffic engineer to synchronize two consecutive intersections' stop lights. The planner claims that if the first light is green the second light will be green with probability 0.75. Assume the probability of finding the first light green is 0.55. What is the probability that any given driver will breeze through both (green) lights?

Let $P(A) = 0.55$ and $P(B|A) = 0.75$. We want to find $P(A \cap B)$.

Conditional Probability, online interactive explanation

Visual Explanations Blog: conditional probability

Conditional Probability, *spam* and *format*

Find $P(\textit{format}|\textit{spam})$.

Conditional Probability, *spam* and *format*

Find $P(\textit{spam}|\textit{format})$.

Motivating Law of Total Probability

Visually, we could sum the probabilities of the intersections of B with each A_j , where all A_1 through A_k are disjoint and $\sum_{i=1}^k P(A_i) = 1$:

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_k \cap B).$$

Law of Total Probability

The law of total probability combines this idea with the formula from conditional probability.

Law of Total Probability

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_k)P(B|A_k)$$

Note that all we've done is substitute each term from the previous slide using the fact that $P(A \cap B) = P(A)P(B|A)$.

Law of Total Probability

Suppose further that we know, if the driver approaches a NON-green traffic light first, then the second light will be green with 9% probability. What is the probability that a driver will find the second traffic light green?

Law of Total Probability

We want to find the probability the second traffic light is green.

What do we know?

- ▶ Know

- ▶ probability first light is green, $P(A) = 0.55$.
- ▶ probability first light is not green, $1 - P(A) = P(A^c) = 0.45$
- ▶ probability second light is green, given first light is green, $P(B|A) = 0.75$
- ▶ probability second light is green, given first light is not green, $P(B|A^c) = 0.09$

- ▶ Want

- ▶ $P(B)$

Law of Total Probability

Find the probability that the first traffic light is green **given that** the second traffic light is green.

► Know

► $P(A) = 0.55.$

► $P(A^c) = 0.45$

► $P(B|A) = 0.75$

► $P(B|A^c) = 0.09$

► $P(B) = 0.453$

► Want

► $P(A|B)$

Bayes' Theorem

Conditional probability combined with the law of total probability, written in all generality (and that we just used), is known as Bayes' Theorem.

Bayes' Theorem

$$P(A_j|B) = \frac{P(A_j)P(B|A_j)}{\sum_{i=1}^k P(A_i)P(B|A_i)}$$

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Independence, intuition

The intuition of independence is best seen mathematically via conditional probability,

$$P(B|A) = P(B).$$

That is, the probability of B does not depend on A .

Independence

The formal definition of independence is a condition that must be checked to state that two events are independent.

pairwise independence

Two events A and B are said to be **pairwise independent** if

$$P(A \cap B) = P(A)P(B).$$

Independence, example

Randomly draw a card from a standard 52 card deck. Let A be the event that the card is a five and B be the event that the card is a spade. Are the events A and B independent?

$P(A \cap B) = 1/52$. Next $P(A) = 4/52$ and $P(B) = 13/52$ gives $P(A)P(B) = 1/52$. Therefore $P(A \cap B) = P(A)P(B)$, so the events are independent.

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Association Rules, motivation

From the following table, calculate and interpret $P(\text{tea})$, $P(\text{coffee})$, $P(\text{coffee} \cap \text{tea})$, $P(\text{coffee}|\text{tea})$.

	coffee	coffee ^c	
tea	150	50	200
tea ^c	650	150	800
	800	200	1000

Table 2: Survey of coffee and tea drinkers¹.

¹Pang-Ning Tan, Michael Steinbach, and Vipin Kumar. Introduction to data mining, 2005.

Association Rules, notation

To investigate an association rule between two sets, here coffee and tea, we write

$$\{tea\} \rightarrow \{coffee\}.$$

Despite the “fancy” words used in *market basket analysis*, I encourage you to think of these concepts in terms of probability, conditional probability, and independence.

Association Rules, definitions

probability (support)

The probability (support) of an association rule $\{B\} \rightarrow \{A\}$ is the proportion of times the rule is present within the dataset:

$$P(A \cap B)$$

conditional probability (confidence)

The conditional probability (confidence) of an association rule $\{B\} \rightarrow \{A\}$ is the support of A and B divided by the support of B :

$$P(A|B)$$

lift

The lift of an association rule $\{B\} \rightarrow \{A\}$ is the confidence (of the rule) divided by the probability of A :

$$P(A|B)/P(A)$$

Association Rules, example

From the Table 2, we have

$$P(\text{tea}) = .2$$

$$P(\text{coffee}) = .8$$

$$P(\text{coffee} \cap \text{tea}) = .15$$

$$P(\text{coffee}|\text{tea}) = .75$$

What is the lift of $\{\text{tea}\} \rightarrow \{\text{coffee}\}$?

Association Rules, justification

Consider two events, A, B and recall independence. What condition must hold for A, B to be independent?

Association Rules, interpreting lift

Lift of the rule $\{B\} \rightarrow \{A\}$ is

$$\frac{P(A \cap B)}{P(A)P(B)}.$$

Interpret

- ▶ lift = 1,
- ▶ lift > 1,
- ▶ lift < 1.

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Take Away

- ▶ Probability is the basis for much of Statistics
- ▶ In the weeks ahead, we'll go from statements like $E = \{\text{flipped head}\}$ to ones like $E = \{X > 3\}$
- ▶ Association rules are new and pretty popular in machine learning:
 - ▶ check out http://michael.hahsler.net/research/arules_RUG_2015/demo/

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Pang-Ning Tan, Michael Steinbach, and Vipin Kumar.
Introduction to data mining, 2005.

Adapted from notes by Dr. Roualdes.